

DIS - DecisionDraft- V1

1. Ingestion

Payload:

GovDirect :

HO-SW-051.json

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{  
  "source_application_id": "HO-SW-2026-76350499",  
  "source_reference": "GV-REF-WHFWJLZH",  
  "source_channel": "govdirect",  
  "visa_type": "skilled-worker",  
  "country_code": "GB",  
  "submitted_at": "2026-03-29T09:42:18Z",  
  "caseworker_id": "HO-CW-3639",  
  "applicant": {  
    "email": "anjali.bose@gmail.com",  
    "first_name": "Anjali",  
    "last_name": "Bose",  
    "date_of_birth": "1977-09-30",  
    "nationality_code": "IN",  
    "gender": "F"  
  },  
  "passport_data": {  
    "number": "C5226129",  
    "issuing_country": "IN",
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[illegible]

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"lastName": "Bose",

"dateOfBirth": "1977-09-30",

"nationality": "IN",

"maritalStatus": "divorced",

"spouseName": null

},

"employment": {

  "status": "employed",

  "employerName": "Meridian Technologies Ltd",

  "jobTitle": "Programmers and Software Development Professionals",

  "annualIncome": 60590,

  "cosReferenceNumber": "COS-2026-2997902",

  "socCode": "2134",

  "startDate": "2026-06-10",

  "salaryFrequency": "ANNUAL",

  "hoursPerWeek": 38.0,

  "isNewEntrant": false,

  "newEntrantReason": null

},

"fees": {

  "ihsReferenceNumber": "IHS-2026-5022-1266",

  "visaFeeReference": "VF-2026-SW-573179",

  "paymentConfirmed": true

},

"englishLanguage": {
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"exemptNationality": false,  
  
"testType": "IELTS_ACADEMIC",  
  
"testReferenceNumber": "26IN923674BOS587U",  
  
"overallScore": 8.0,  
  
"speakingScore": 7.0,  
  
"listeningScore": 7.5,  
  
"readingScore": 8.0,  
  
"writingScore": 8.5,  
  
"testDate": "2026-01-14"  
  
},  
  
"travelHistory": {  
  
  "previousUkVisas": [],  
  
  "previousUkRefusals": false,  
  
  "previousDeportations": false,  
  
  "countriesVisitedLast5Years": [  
  
    "IN",  
  
    "GB",  
  
    "SG"  
  
  ]  
  
},  
  
"criminalRecord": {  
  
  "hasConvictions": false,  
  
  "hasPendingCharges": false,  
  
  "hasCivilJudgments": false  
  
},
```

```
"healthDeclaration": {  
  
  "hasTbCertificate": true,  
  
  "hasHealthConditions": false,  
  
  "requiresAccessibility": false,  
  
  "tbCertificateClinicName": "IOM Migration Health Assessment Centre, New Delhi",  
  
  "tbCertificateClinicCountry": "IN"  
  
}
```

```
},
```

```
"documents": [  
  

```

```
{
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  "document_id": "ho-doc-001-6c4ae2fc",
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  "type": "PASSPORT",
```

```
  "filename": "passport_bio_page.jpg",
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  "signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/PASSPORT/passport\_bio\_page.jpg?sig=HMAC\_GOV\_SIGNATURE\_REDACTED&exp=2026-03-29T01:00:00Z",
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  "mime_type": "image/jpeg",
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  "size_bytes": 2835656,
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  "uploaded_at": "2026-03-28T22:00:00Z",
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  "scanned_by": "HO-CW-3639",
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```
  "scan_location": "Home Office Casework Hub — Glasgow"
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},
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{
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  "document_id": "ho-doc-003-0d10e00d",
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  "type": "EMPLOYMENT_LETTER",
```

```
  "filename": "meridian_technologie_employment_confirmation.pdf",
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"signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/EMPLOYMENT_LETTER/meridian_technologie_employment_confirmation.pdf?sig=HMAC_GOV_SIGNATURE_REDACTED&exp=2026-03-29T01:00:00Z",

"mime_type": "application/pdf",

"size_bytes": 497120,

"uploaded_at": "2026-03-28T22:05:00Z",

"scanned_by": "HO-CW-3639",

"scan_location": "Home Office Casework Hub — Glasgow"

},

{

"document_id": "ho-doc-004-494871b1",

"type": "BANK_STATEMENT",

"filename": "sbi_statement_jan_2026.pdf",

"signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/BANK_STATEMENT/sbi_statement_jan_2026.pdf?sig=HMAC_GOV_SIGNATURE_REDACTED&exp=2026-03-29T01:00:00Z",

"mime_type": "application/pdf",

"size_bytes": 1052805,

"uploaded_at": "2026-03-28T22:09:00Z",

"scanned_by": "HO-CW-3639",

"scan_location": "Home Office Casework Hub — Glasgow"

},

{

"document_id": "ho-doc-005-52a6f1b8",

"type": "BANK_STATEMENT",

"filename": "sbi_statement_feb_2026.pdf",

"signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/BANK_STATEMENT/sbi_statement_feb_2026.pdf?sig=HMAC_GOV_SIGNATURE_REDACTED&exp=2026-03-29T01:00:00Z",

"mime_type": "application/pdf",

"size_bytes": 955162,

"uploaded_at": "2026-03-28T22:10:00Z",

"scanned_by": "HO-CW-3639",

"scan_location": "Home Office Casework Hub — Glasgow"

},

{

"document_id": "ho-doc-006-5dac9408",

"type": "DEGREE_CERTIFICATE",

"filename": "degree_bsc_meridian_t.pdf",

"signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/DEGREE_CERTIFICATE/degree_bsc_meridian_t.pdf?sig=HMAC_GOV_SIGNATURE_REDACTED&exp=2026-03-29T01:00:00Z",

"mime_type": "application/pdf",

"size_bytes": 1855396,

"uploaded_at": "2026-03-28T22:16:00Z",

"scanned_by": "HO-CW-3639",

"scan_location": "Home Office Casework Hub — Glasgow"

},

{

"document_id": "ho-doc-007-1c05bf0a",

"type": "IELTS_CERTIFICATE",

"filename": "pte_result_certificate.pdf",

"signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/IELTS_CERTIFICATE/pte_result_certificate.pdf?sig=HMAC_GOV_SIGNATURE_REDACTED&exp=2026-03-29T01:00:00Z",

"mime_type": "application/pdf",

"size_bytes": 433426,

"uploaded_at": "2026-03-28T22:20:00Z",

"scanned_by": "HO-CW-3639",

"scan_location": "Home Office Casework Hub — Glasgow"

},

{

"document_id": "ho-doc-008-b1029319",

"type": "PAYSLIPS",

"filename": "meridian_technologie_payslips_dec_feb_2026.pdf",

"signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/PAYSLIPS/meridian_technologie_payslips_dec_feb_2026.pdf?sig=HMAC_GOV_SIGNATURE_REDACTED&exp=2026-03-29T01:00:00Z",

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"size_bytes": 688100,

"uploaded_at": "2026-03-28T22:18:00Z",

"scanned_by": "HO-CW-3639",

"scan_location": "Home Office Casework Hub — Glasgow"

},

{

"document_id": "ho-doc-009-34e2ffd5",

"type": "TB_CERTIFICATE",

"filename": "tb_certificate_iom.pdf",

"signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/TB_CERTIFICATE/tb_certificate_iom.pdf?sig=HMAC_GOV_SIGNATURE_REDACTED&exp=2026-03-29T01:00:00Z",

"mime_type": "application/pdf",

"size_bytes": 271331,

"uploaded_at": "2026-03-28T22:24:00Z",

"scanned_by": "HO-CW-3639",

"scan_location": "Home Office Casework Hub — Glasgow"

},

{

"document_id": "ho-doc-010-64d47e1f",

"type": "PHOTO",

"filename": "bose_passport_photo.jpg",

"signed_url": "https://gov-doc-store.homeoffice.gov.uk/applications/HO-SW-2026-76350499/PHOTO/bose_passport_photo.jpg?sig=HMAC_GOV_SIGNATURE_REDACTED&exp=2026-03-29T01:00:00Z",

"mime_type": "image/jpeg",

"size_bytes": 1851546,

"uploaded_at": "2026-03-28T22:27:00Z",

"scanned_by": "HO-CW-3639",

"scan_location": "Home Office Casework Hub — Glasgow"

}

],

"callback_url": "<https://casework-api.homeoffice.gov.uk/api/v1/dis-decisions/HO-SW-2026-76350499>"

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VisaKey :

VK-SW-001.json

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{  
  "source_application_id": "c33f4584-b23b-c1d8-493c-d01609de8895",  
  "source_reference": "VK-J3XG35B6",  
  "source_channel": "visakey",  
  "visa_type": "skilled-worker",  
  "country_code": "GB",  
  "submitted_at": "2026-03-29T09:42:18Z",  
  "applicant": {  
    "user_id": "user_BKxN28J4W8BzId",  
    "email": "arjun.reddy@outlook.com",  
    "first_name": "Arjun",  
    "last_name": "Reddy",  
    "date_of_birth": "1979-01-16",  
    "nationality_code": "IN",  
    "gender": "M"  
  },  
  "passport_data": {  
    "number": "L1797908",  
    "issuing_country": "IN",  
    "nationality": "IN",  
    "full_name": "REDDY ARJUN",  
    "date_of_birth": "1979-01-16",
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[illegible]

```
"employerName": "Meridian Technologies Ltd",  
"jobTitle": "Programmers and Software Development Professionals",  
"annualIncome": 55861,  
"cosReferenceNumber": "COS-2026-7247410",  
"socCode": "2134",  
"startDate": "2026-06-06",  
"salaryFrequency": "ANNUAL",  
"hoursPerWeek": 37.5,  
"isNewEntrant": false,  
"newEntrantReason": null  
},  
"fees": {  
  "ihsReferenceNumber": "IHS-2026-5323-7949",  
  "visaFeeReference": "VF-2026-SW-573685",  
  "paymentConfirmed": true  
},  
"englishLanguage": {  
  "exemptNationality": false,  
  "testType": "IELTS_GENERAL",  
  "testReferenceNumber": "26IN013247RED136Z",  
  "overallScore": 6.5,  
  "speakingScore": 7.0,  
  "listeningScore": 6.0,  
  "readingScore": 7.5,  
  "writingScore": 5.5,
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"testDate": "2025-07-18"

},

"travelHistory": {

  "previousUkVisas": [

    {

      "type": "Standard Visitor Visa",

      "from": "2022-07-10",

      "to": "2022-09-20",

      "reference": "VIS-2022-SVV-234956"

    }

  ],

  "previousUkRefusals": false,

  "previousDeportations": false,

  "countriesVisitedLast5Years": [

    "IN",

    "GB",

    "AE"

  ]

},

"criminalRecord": {

  "hasConvictions": false,

  "hasPendingCharges": false,

  "hasCivilJudgments": false

},

"healthDeclaration": {
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```
"hasTbCertificate": true,

"hasHealthConditions": false,

"requiresAccessibility": false,

"tbCertificateClinicName": "IOM Migration Health Assessment Centre, Mumbai",

"tbCertificateClinicCountry": "IN"

},

},

"documents": [

{

"document_id": "44a3f944-9766-8966-e05a-dc180cd32c52",

"type": "PASSPORT",

"filename": "passport_bio_page.jpg",

"gcs_path": "user_BKxN28J4W8Bzld/c33f4584/PASSPORT/1774735020-

passport_bio_page.jpg",

"signed_url": "https://storage.googleapis.com/visa-key-documents/user\_BKxN28J4W8Bzld/c33f4584/PASSPORT/1774735020-passport\_bio\_page.jpg?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visa-key-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE\_SIGNATURE\_REDACTED",

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"size_bytes": 2381345,

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{

"document_id": "fe4c06d4-f81b-d0ee-1202-87d69abf39b8",

"type": "EMPLOYMENT_LETTER",
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"filename": "meridian_technologie_employment_confirmation.pdf",

"gcs_path": "user_BKxN28J4W8Bzld/c33f4584/EMPLOYMENT_LETTER/1774735440-meridian_technologie_employment_confirmation.pdf",

"signed_url": "https://storage.googleapis.com/visakey-documents/user_BKxN28J4W8Bzld/c33f4584/EMPLOYMENT_LETTER/1774735440-meridian_technologie_employment_confirmation.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visakey-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE_SIGNATURE_REDACTED",

"mime_type": "application/pdf",

"size_bytes": 348921,

"uploaded_at": "2026-03-28T22:04:00Z"

},

{

"document_id": "4c0897ce-1d01-59fd-a9a3-d6238d4de4f9",

"type": "BANK_STATEMENT",

"filename": "sbi_statement_dec_2025.pdf",

"gcs_path": "user_BKxN28J4W8Bzld/c33f4584/BANK_STATEMENT/1774735440-sbi_statement_dec_2025.pdf",

"signed_url": "https://storage.googleapis.com/visakey-documents/user_BKxN28J4W8Bzld/c33f4584/BANK_STATEMENT/1774735440-sbi_statement_dec_2025.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visakey-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE_SIGNATURE_REDACTED",

"mime_type": "application/pdf",

"size_bytes": 955273,

"uploaded_at": "2026-03-28T22:04:00Z"

},

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{  
  "document_id": "ba83f02a-c775-3887-e63c-e18b7eb39bc1",  
  "type": "BANK_STATEMENT",  
  "filename": "sbi_statement_jan_2026.pdf",  
  "gcs_path": "user_BKxN28J4W8Bzld/c33f4584/BANK_STATEMENT/1774735440-sbi_statement_jan_2026.pdf",  
  "signed_url": "https://storage.googleapis.com/visakey-documents/user\_BKxN28J4W8Bzld/c33f4584/BANK\_STATEMENT/1774735440-sbi\_statement\_jan\_2026.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visakey-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE\_SIGNATURE\_REDACTED",  
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  "size_bytes": 1073810,  
  "uploaded_at": "2026-03-28T22:04:00Z"  
},  
  
{  
  "document_id": "6f1a311e-b715-e9a1-5c8f-fa6ef489e08b",  
  "type": "BANK_STATEMENT",  
  "filename": "sbi_statement_feb_2026.pdf",  
  "gcs_path": "user_BKxN28J4W8Bzld/c33f4584/BANK_STATEMENT/1774735500-sbi_statement_feb_2026.pdf",  
  "signed_url": "https://storage.googleapis.com/visakey-documents/user\_BKxN28J4W8Bzld/c33f4584/BANK\_STATEMENT/1774735500-sbi\_statement\_feb\_2026.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visakey-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE\_SIGNATURE\_REDACTED",  
  "mime_type": "application/pdf",
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"size_bytes": 1106381,

"uploaded_at": "2026-03-28T22:05:00Z"

},

{

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  "type": "DEGREE_CERTIFICATE",

  "filename": "degree_bsc_meridian_t.pdf",

  "gcs_path": "user_BKxN28J4W8Bzld/c33f4584/DEGREE_CERTIFICATE/1774736040-degree_bsc_meridian_t.pdf",

  "signed_url": "https://storage.googleapis.com/visakey-documents/user\_BKxN28J4W8Bzld/c33f4584/DEGREE\_CERTIFICATE/1774736040-degree\_bsc\_meridian\_t.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visakey-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE\_SIGNATURE\_REDACTED",

  "mime_type": "application/pdf",

  "size_bytes": 2212247,

  "uploaded_at": "2026-03-28T22:14:00Z"

},

{

  "document_id": "5badd3cc-1569-f64b-c60e-43ae9a1394ff",

  "type": "IELTS_CERTIFICATE",

  "filename": "pte_result_certificate.pdf",

  "gcs_path": "user_BKxN28J4W8Bzld/c33f4584/IELTS_CERTIFICATE/1774735980-pte_result_certificate.pdf",

  "signed_url": "https://storage.googleapis.com/visakey-documents/user\_BKxN28J4W8Bzld/c33f4584/IELTS\_CERTIFICATE/1774735980-pte\_result\_certificate.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-
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"mime_type": "application/pdf",

"size_bytes": 726499,

"uploaded_at": "2026-03-28T22:13:00Z"

},

{

"document_id": "df9e9526-1613-064f-a51c-6d2a208d57e3",

"type": "PAYSLIPS",

"filename": "meridian_technologie_payslips_dec_feb_2026.pdf",

"gcs_path": "user_BKxN28J4W8Bzld/c33f4584/PAYSLIPS/1774736280-meridian_technologie_payslips_dec_feb_2026.pdf",

"signed_url": "https://storage.googleapis.com/visa-key-documents/user_BKxN28J4W8Bzld/c33f4584/PAYSLIPS/1774736280-meridian_technologie_payslips_dec_feb_2026.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visa-key-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE_SIGNATURE_REDACTED",

"mime_type": "application/pdf",

"size_bytes": 979267,

"uploaded_at": "2026-03-28T22:18:00Z"

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"type": "TB_CERTIFICATE",

"filename": "tb_certificate_iom.pdf",

"gcs_path": "user_BKxN28J4W8Bzld/c33f4584/TB_CERTIFICATE/1774736280-tb_certificate_iom.pdf",

"signed_url": "https://storage.googleapis.com/visa-key-documents/user_BKxN28J4W8Bzld/c33f4584/TB_CERTIFICATE/1774736280-tb_certificate_iom.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visa-key-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE_SIGNATURE_REDACTED",

"mime_type": "application/pdf",

"size_bytes": 388645,

"uploaded_at": "2026-03-28T22:18:00Z"

},

{

"document_id": "b0a84cd0-1be3-8ac1-4eb9-d5a9ceb64bfe",

"type": "PROOF_OF_ADDRESS",

"filename": "council_tax_bill_2026.pdf",

"gcs_path": "user_BKxN28J4W8Bzld/c33f4584/PROOF_OF_ADDRESS/1774741200-council_tax_bill_2026.pdf",

"signed_url": "https://storage.googleapis.com/visa-key-documents/user_BKxN28J4W8Bzld/c33f4584/PROOF_OF_ADDRESS/1774741200-council_tax_bill_2026.pdf?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visa-key-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE_SIGNATURE_REDACTED",

"mime_type": "application/pdf",

"size_bytes": 283766,

"uploaded_at": "2026-03-28T23:40:00Z"

},

{

"document_id": "964ee498-716d-78b2-6cbc-093c890221e8",

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"type": "PHOTO",

"filename": "reddy_passport_photo.jpg",

"gcs_path": "user_BKxN28J4W8Bzld/c33f4584/PHOTO/1774736760-
reddy_passport_photo.jpg",

"signed_url": "https://storage.googleapis.com/visa-key-documents/user\_BKxN28J4W8Bzld/c33f4584/PHOTO/1774736760-reddy\_passport\_photo.jpg?X-Goog-Algorithm=GOOG4-RSA-SHA256&X-Goog-Credential=visa-key-api%40prj-dev-dis-9666.iam.gserviceaccount.com&X-Goog-Date=20260329T000000Z&X-Goog-Expires=900&X-Goog-Signature=EXAMPLE\_SIGNATURE\_REDACTED",

"mime_type": "image/jpeg",

"size_bytes": 1460103,

"uploaded_at": "2026-03-28T22:26:00Z"

},

],

"eligibility_score": {

"total": 40,

"breakdown": {

"passport": 32,

"residence": 0,

"existingVisas": 2,

"bonus": 6

},

"passport_country": "IN",

"residence_country": "IN"

},

"callback_url": "https://visa-key-api-977813303563.europe-west2.run.app/api/webhooks/dis-decision"
```

}

2. Data Classification / Extraction:

Name of Document	Tier	Processing Type	Fields to be Extracted	Storage Strategy
Passport	Tier 1	DOC AI	document_number, surname, given_names, full_name, date_of_birth, nationality, sex, issue_date, expiry_date, issuing_country, country_code, document_type, place_of_birth, place_of_issue, mrz_line_1, mrz_line_2,	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields

			photo_hash, has_signatur e, document_a uthenticity_s core, fraud_flags, fraud_signals , confidence_s core	
Bank Statement	Tier 1	DOC AI	account_hold er_name, account_nu mber, sort_code, micr_code, ifsc_code, customer_id, account_curr ency, product_nam e, branch_nam e, statement_p eriod, opening_bala nce, closing_bala nce, confidence_s core	Stored in Postgres document_ex tractions.raw _extraction (JSONB) → Normalized into normalized_fi elds
National Identity Card	Tier 1	DOC AI	document_n umber, given_names	Stored in Postgres document_ex

			, surname, full_name, date_of_birth , nationality, gender, issue_date, expiry_date, issuing_auth ority, has_photo, confidence_s core	tractions.raw _extraction (JSONB) → Normalized into normalized_fi elds
Biometric Residence Permit (BRP)	Tier 1	DOC AI	document_n umber, given_names , surname, full_name, date_of_birth , nationality, gender, issue_date, expiry_date, issuing_auth ority, has_photo, confidence_s core	Stored in Postgres document_ex tractions.raw _extraction (JSONB) → Normalized into normalized_fi elds
Employment Letter / Contract	Tier 2	TBD	employer_na me, job_title, start_date, salary_amou nt, salary_freque ncy, employment_ type,	Stored in Postgres document_ex tractions.raw _extraction (JSONB) → Normalized into

			hours_per_week, employer_address, signatory_name, signatory_position, letter_date	normalized_fields
Payslips	Tier 2	TBD	employer_name, employee_name, pay_period_start, pay_period_end, gross_pay, net_pay, tax_deducted, ni_deducted, pay_frequency, payslip_date, employee_number	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields
P60 / Tax Document	Tier 2	TBD	tax_year, employer_name, employee_name, ni_number, total_pay_in_year, total_tax_in_y	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into

			ear, total_ni_in_year, employer_pay_reference	normalized_fields
Educational Certificate	Tier 2	TBD	institution_name, candidate_name, qualification_title, qualification_level, subject, award_date, classification , country_of_institution, naric_reference	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields
COS Letter	Tier 2	TBD	cas_number, institution_name, institution_licence_number, student_name, date_of_birth, course_title, course_level, course_start_date, course_end_date,	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields

			tuition_fee, fees_paid, cas_assigned_date, cas_expiry_date	
Endorsement Letter	Tier 2	TBD	endorsing_body, visa_route, applicant_name, endorsement_reference, endorsement_date, endorsement_expiry_date, endorsement_category, signatory_name, signatory_position	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields
Marriage Certificate	Tier 2	TBD	spouse_1_full_name, spouse_1_date_of_birth, spouse_1_nationality, spouse_2_full_name, spouse_2_date_of_birth, spouse_2_nationality, marriage_date	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields

			e, marriage_place, certificate_number, issuing_authority, country_of_issue	
Birth Certificate	Tier 2	TBD	child_full_name, date_of_birth, place_of_birth, sex, father_full_name, mother_full_name, certificate_number, registration_date, issuing_authority, country_of_issue	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields
Persecution Evidence	Tier 2	TBD	document_type, subject_name, date_of_document, issuing_authority_or_source,	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into

			country_of_origin, persecution_grounds, reference_number, summary	normalized_fields
Witness Statement	Tier 2	TBD	witness_full_name, witness_date_of_birth, witness_nationality, witness_address, relationship_to_applicant, statement_date, statement_reference, subject_of_statement, signature_present, witnessed_by	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields
Business Plan	Tier 2	TBD	business_name, applicant_name, business_sector, business_description, investment_a	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into

			mount, funding_source, projected_revenue_year_1, jobs_to_be_created, innovation_statement, document_date, endorsing_body	normalized_fields
Funding Evidence	Tier 2	TBD	document_type, fund_holder_name, funding_amount, funding_currency, funding_source, issuing_institution, document_date, funds_available_from, account_number_last4, reference_number	Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields
Utility Bill	Tier 2	TBD	account_holder_name, service_addr	Stored in Postgres document_ex

			ess, utility_type, supplier_name, bill_date, billing_period_start, billing_period_end, account_number, amount_due	tractions.raw _extraction (JSONB) → Normalized into normalized_fields
Tenancy Agreement	Tier 2	TBD	tenant_full_name, landlord_full_name, property_address, tenancy_start_date, tenancy_end_date, tenancy_type , monthly_rent, deposit_amount, agreement_date, managing_agent	Stored in Postgres document_extractions.raw _extraction (JSONB) → Normalized into normalized_fields
Police Certificate	Tier 2	TBD	subject_full_name, date_of_birth, nationality, issuing_authority,	Stored in Postgres document_extractions.raw _extraction (JSONB) →

			issuing_coun try, certificate_n umber, issue_date, expiry_date, criminal_reco rd_disclosed, certificate_ty pe	Normalized into normalized_fi elds
Medical Certificate	Tier 2	TBD	patient_full_n ame, date_of_birth , certificate_ty pe, issuing_clinic _or_hospital, issuing_coun try, examining_d octor_name, issue_date, expiry_date, certificate_n umber, outcome	Stored in Postgres document_ex tractions.raw _extraction (JSONB) → Normalized into normalized_fi elds
IELTS Certificate	Tier 2	TBD	test_type, candidate_na me, date_of_birth , test_date, overall_score , listening_sco re,	Stored in Postgres document_ex tractions.raw _extraction (JSONB) → Normalized into

			reading_score, writing_score, , speaking_score, test_report_form_number, cefr_level	normalized_fields
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All extracted data are first stored in the PostgreSQL table **document_extractions**:

- Column **raw_extraction** – jsonb containing the full set of fields as shown in the *Fields to be extracted* column.
- Column **normalized_fields** – jsonb (or a set of dedicated columns) that holds the cleaned / normalized representation of each field for downstream use.

3. External API Checks

External Check	Description	Request Parameters	Response Parameters	Validation Checks	Storage
Reuters Wold Check API	Sanctions & PEP screening Sanctions, PEP	request_id, dis_application_id, document_id,	request_id, check_type, check_status,risk_level	Sanctions Match: risk_level == HIGH PEP Match:	external_checks

(Politically Exposed Person), and adverse media screening. Triggered on EVERY application regardless of visa type. This is a universal check.	check_type, api_version, requested_at, full_name, date_of_birth, nationality, passport_number, gender, secondary_names, country_of_residence	vel, confidence_score, flags, screening_id, matches_found, categories_checked {"SANCTIONS", "PEP", "ADVERSE_MEDIA" , "LAW_ENFORCEMENT"}, lists_checked{"OFSI", "UN_SC", "EU_SANCTIONS", "US_OFAFC", "INTERPOL_RED_NOTICE"} , match_details, screening_date, response	risk_level == MEDIUM, flag PEP_MATCH No match: risk_level == NONE API timeout/error: check_status == ERROR/TIMEOUT
---	---	---	---

			d_at, response_ time_ms		
Interpol Check	Stolen/Lost travel doc check Checks passport document number against Interpol's Stolen and Lost Travel Documents (SLTD) database. Triggered whenever a passport document is detected in the application package (document _type == PASSPORT).	request_id, dis_application_id, document_id, check_type, api_version, requested_at, document_number, document_type,issuing_count ry, holder_name, date_of_birth, expiry_date	request_id, check_type, check_status,risk_level, confidence_score, flags,sltd_reference, database_version, document_number_ checked, is_stolen, is_lost, is_revoked, is_invalid, report_details, responded_at, response_time_ms	is_stolen: True / False is_lost: True / False is_Revoked: True / False is_invalid: True / False All flags false: True / False	external_checks
Passport Verification	MRZ cross-check & authenticity	request_id, dis_application_id, document_id,	request_id, check_type, check_status,risk_level	MRZ mismatch: overall_match == false Low	external_checks

Cross-references extracted MRZ data against issuing authority records. Verifies document authenticity, confirms identity details, and checks document validity status. Triggered when document_type == PASSPORT.	check_type, api_version, requested_at, document_number, document_type, issuing_count, ry, holder_name, date_of_birth, expiry_date, mrz_line_1, mrz_line_2, issuing_country, extracted_surname, extracted_given_names, extracted_dob, extracted_expiry, extracted_gender, document	vel, confidence_score, flags, verification_id, mrz_cross_reference, overall_match, field_matches, document_number, surname, given_names, date_of_birth, expiry_date, gender, nationality, check_digit_valid, authenticity, document_status, responded_at, response_time_ms	authenticity: authenticity_score < 0.50 Borderline authenticity: score >= 0.50 AND < 0.80 Document cancelled : is_cancelled == true All checks pass: match == true, score >= 0.80, status == V ALID
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		_image_hash			
Border Control	Entry/exit history Returns travel history for an applicant — UK entry/exit records. Used by Drools to validate claimed residence periods, check for overstays, and verify travel patterns consistent with visa type. Triggered on every application .	request_id, dis_application_id, document_id, check_type, api_version, requested_at, document_number(Passport_Number), full_name, date_of_birth, nationality, lookup_period_months	request_id, check_type, check_status, risk_level, confidence_score, flags, border_ref, records_founded, has_overstay, has_deportation, has_refusal_at_border, travel_records {Entry / Exit }, current_immigration_status, responded_at, response_time_ms	Previous overstay: has_overstay == true Previous deportation: has_deportation == true Border refusal: has_refusal_at_border == true Currently in UK (overstay?): is_currently_in_uk && leave_expired Clean history: All flags false	external_checks
Device & Ip Check	Submission fraud	request_id,	request_id,	Known fraud	external_checks

signals	dis_applic	check_type,	IP:is_known_fraud_ip == true
Evaluates fraud risk signals from the submission device and IP address.	ation_id,	check_status,	Tor exit node:
Catches VPN usage, known fraud IP ranges, impossible travel, and device fingerprint anomalies.	document_id,	risk_level,	is_tor == true
Triggered on every submission (not per document — per application submission event).	check_type,	confidence_score,	VPN + geo mismatch:
The document_id field is NULL for this check.	e,	flags,	is_vpn && geo_mismatch
	api_version,	risk_ref,	Impossible travel: impossible_travel == true
	n,	ip_analyses,	Rapid resubmission: is_rapid_resubmission == true
	requested_at,	ip_addresses,	Clean submission: No flags triggered
	ip_addresses,	ip_addresses,	
	device_fingerprint,	ip_countries,	
	user_agent,	ip_city,	
	t,	is_vpn,	
	geo_location,	is_tor,	
	submission_timestamp,	is_proxy,	
	n_timestamp,	is_datacenter,	
	session_id,	is_known_fraud_ip,	
	d,	isp,	
	previous_submission_count	device_analysis,	
		device_fingerprint_known,	
		device_first_seen,	
		multiple_accounts_same_device,	
		device_trust_score,	
		geo_analy	

			sis, submissio n_velocity , submissio ns_last_2 4h, submissio ns_last_7 d, is_rapid_r esubmissi on, responde d_at, response_ time_ms		
Email & Phone Reputation	Contact detail verificatio n Verifies applicant contact details — checks whether email and phone number are legitimate, recently created (disposabl e), or	request_i d, dis_applic ation_id, document _id, check_typ e, api_versio n, requested _at, email_add ress, phone_nu mber, phone_co untry, applicant_ name	request_i d, check_typ e, check_sta tus,risk_le vel, confidenc e_score, flags,repu tation_ref, email_ana lysis {email_ad dress, is_valid_fo rmat, is_deliver able, is_dispos	Disposabl e email: is_disposa ble == true Known fraud email/pho ne:is_kno wn_fraud_ * == true VOIP / Virtual number: is_voip is_virtual Undelivera ble email: is_delivera ble ==	external_c hecks

	associated with known fraud. Triggered on every application using contact details from the application form.	able, is_free_provider, domain_age_days, first_seen_date, breach_count, is_known_fraud_email, name_email_correlation}, phone_analysis{phone_number, is_valid, line_type, carrier, is_voip, is_virtual, is_prepaid , phone_country, is_known_fraud_phone, first_seen_date}, responded_at, response_time_ms	false Phone country mismatch: phone_country != claimed nationality All clean:No flags triggered	
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4. Drools

Rules		Input Data	Source Input	Require Reference Data	Reference Json Name	Ref Field Name	Remarks	Storage
universal/passport_rules.drl								
RULE-U01: Passport Validity (expiry ≥ 6 months from travel date)		expiry_date	EXTRACTI ON(passport) - document_ extrac tions table	No				rule_results
universal/biometric_rules.drl								
	RULE-U02: Biometric Verification (face match ≥ 0.85, liveness = PASS	face_match , liveness and MRZ	Submission Payload(Visa Key)EXTRA CTION(pas sport) - document_ extrac	No				rule_results

	ED, MRZ = PASS ED)		tions table					
	univer sal/sa nction s_rule s.drl							
	RULE- U03: Sancti ons Scree ning (World - Check HIGH = imme diate block, MEDI UM = manu al review)	risk_le vel	Extern al API(W orld- Check)- extern al_che cks table	No				rule_r esults
	univer sal/du plicate _rules. drl							

	RULE-U04: Duplicate Application Check (passport + DOB + visa type + status + 6-month window)						Preety to validation clarification from Open Visa	rule_results
	universal/document_confidence_rules.drl							
	RULE-U05: Document Extraction Confidence (all Tier 1 docs $\geq$ 0.80)	document_type, confidence_score	EXTRACTION(All Docs) - document_extractions table-	No				rule_results

	confid ence)		norma lised_f ields					
	skilled _work er/spo nsors hip_rul es.drl							
	RULE- W01: CoS Validit y (prese nt, assign ed within 3 month s, not used)	cos_st atus,c os_ex piry_d ate	Submi ssion Payloa d	Yes	cos_re gister_ mock.j son	cos_st atus,c os_ex piry_d ate		rule_r esults
	RULE- W02: Spons or Licenc e Status (A- rated, not	cos_st atus,c os_ex piry_d ate	Submi ssion Payloa d	Yes	cos_re gister_ mock.j son	cos_st atus,c os_ex piry_d ate		rule_r esults

	suspe nded/ revoke d)							
	RULE- W15: Start Date Validit y (empl oyme nt start within 3 month s of submi ssion)	emplo yment StartD ate cos_e xpiry_ date	Submi ssion Payloa d	Yes	cos_re gister_ mock.j son	cos_e xpiry_ date		rule_r esults
	skilled_worker/salary_rules.drl							
	RULE- W03: Salary Thres hold — Gener al Option A (≥ £41,70 0 AND ≥ 100%	salary _annu al_nor malise d hours_ per_w eek/w eekly_ hours, socCo de	Submi ssion Payloa d	Yes	soc_g oing_r ates.js on	going_ rate_h ourly	Preety to validat ion clarfic ation from Open Visa	rule_r esults

	going_rate AND hourly ≥ £17.13)	going_rate_hourly						
	RULE-W04: Salary Threshold — New Entrant (≥ £33,400 AND ≥ 100% going_rate)	isNewEntrant or to be calculated (age,) date_of_birth socCode going_rate_hourly	Submission Payload, EXTRACTION (Passport)-document_extractions	Yes	soc_going_rates.js on	going_rate_hourly	Preety to validation clarification from Open Visa	rule_results
	RULE-W06: Immigration Salary List Check (reduced threshold if SOC on ISL)	socCode(payload), soc_code(immigration_salary_list.js on) expiry	Submission payload	Yes	immigration_salary_list.js on	soc_code expiry (to be removed post expiry)	Preety to validation clarification from Open Visa	rule_results

	skilled_worker/eligibility_rules.drl							
	RULE-W05: SOC Code Eligibility (SOC 2020 in Appendix Skilled Occupations Tables 1-3, RQF 6+)	socCode, job_title, job_description	Submission payload	Yes	eligibility_soc_code.s.json	soc_Code	Preety to validate clarification from Open Visa	rule_results
	RULE-W07: Job Skill Level (RQF Level 6 / graduate level minimum)	socCode, job_title, rqf_level	Submission payload	Yes	eligibility_soc_code.s.json	rqf_level		rule_results
	RULE-W08: English	overall_score	EXTRACTION (IELTS	No				rule_results

	Langu age Requir ement (CEFR B2+ effecti ve 8 Janua ry 2026, per HC 395 Appen dix Skilled Worke r SW 3.1)	listeni ng_sc ore readin g_scor e writin g_scor e speaki ng_sc ore	Certifi cate)- docu ment_ extrac tions					
	skilled_worker/financial_rules.drl							
	RULE- W11: Crimin al Recor d Disclo sure (Inter pol check — red notice =						Preety to validat ion clarfic ation from Open Visa	rule_r esults

	escala tion)							
	RULE- W10: TB Test Certifi cate (if nation ality in Appen dix T countr y list)	countr y_cod e(payload) countr ies(tb_ test_c ountri es.js on) certifi cate_t ype certifi cate_n umber issuin g_clini c_or_h ospital exami ning_d octor_ name issue_ date outco me	Submi ssion payload EXTR ACTI ON(M edical Certifi cate)- docu ment_ extrac tions	Yes	tb_tes t_coun tries.js on	countr ies		rule_r esults
	skilled_worker/compliance_rules.drl							
	RULE- W11:						Preety to	rule_r esults

	Criminal Record Disclosure (Interpol check — red notice = escalation)						validate clarification from Open Visa	
	RULE-W12: Previous Immigration Compliance (oversight, refusal, deportation history)	has_oversight has_deportation has_refusal_at_border	External API(Border Control)-external_checks table	No				rule_results
	RULE-W14: Document Fraud Detect	fraud_score	EXTERNAL DOCS)-	No				rule_results

	ion (Visio n AI fraud_ score thresh olds)		docu ment_ extrac tions table					
skilled_worker/completeness_rules.drl								
	RULE- W13: Applic ation Compl etene ss (weigh ted scorin g, 0- 100)	compl etene ss_sc ore	EXTE RACTI ON(AL L DOCS) - docu ment_ extrac tions table	No				rule_r esults

5. OPA

Policy Name	Severity (Hard/Soft)	Descripti on	Source Type	Input Paramet ers	Output Result	Storage Location
OPA-H01 (sanctio ns.rego)	Hard	Blocks applicati ons with confirme d sanction s	Reuters World- Check API Border Control API	worldch eck.risk_ level == HIGH; check_st atus == ERROR/	result := { "policy_i d": "OPA- H01",	opa_res ults

		matches or critical World- Check results	TIMEOUT; border_control.de tails.has_deportati on == true [Review Required] "result": "BLOCK", "reason": "World check failed or timed out — fail- closed", "data_source": "REUTERS_WOR LDCHECK", "evaluated_at": input.metadata.ti mestamp,	
--	--	---	--	--

```
} if {  
  
    input.ext  
    ernal_ch  
    ecks.wor  
    ldcheck.  
    check_st  
    atus in  
    ["ERROR  
    ",  
    "TIMEO  
    UT"]  
} else :=  
{  
  
    "policy_i  
    d":  
    "OPA-  
    H01",  
  
    "policy_n  
    ame":  
    "Sanctio  
    ns_Worl  
    dCheck_  
    HardBlo  
    ck",  
    "tier":  
    "HARD",  
  
    "result":  
    "BLOCK  
    ",  
  
    "reason"  
    : "Border  
    control
```

check
failed or
timed
out —
fail-
closed",

"data_so
urce":
"BORDE
R_CONT
ROL",

"evaluat
ed_at":
input.me
tadata.ti
mestam
p,
} if {

input.ext
ernal_ch
ecks.bor
der_cont
rol.check
_status
in
["ERROR
",
"TIMEO
UT"]
} else :=
{

"policy_i
d":

"OPA-H01",
"policy_name":
"Sanctions_WorldCheck_HardBlock",
"tier":
"HARD",
"result":
"BLOCK",
"reason":
: "High-risk sanctions match found in World-Check",
"data_source":
"REUTERS_WORLDCHECK",
"evaluated_at":
input.metadata.timestamp

```
mestam
p,
} if {

input.ext
ernal_ch
ecks.wor
ldcheck.
risk_leve
l ==
"HIGH"
} else :=
{

"policy_i
d":
"OPA-
H01",

"policy_n
ame":
"Sanctio
ns_Worl
dCheck_
HardBlo
ck",
    "tier":
"HARD",

"result":
"REVIE
W_REQU
IRED",

"reason"
:
"Deporta
```

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	history	
	detected	
	—	
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	casewor	
	ker	
	review",	
	"data_so	
	urce":	
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	p,	
	} if {	
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	} else :=	
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	"policy_i	
	d":	
	"OPA-	
	H01",	
	"policy_n	
	ame":	
	"Sanctio	
	ns_Worl	

					<pre>dCheck_ HardBlo ck", "tier": "HARD", "result": "PASS", "reason" : "No sanction s match or adverse border control history", "data_so urce": "REUTE RS_WOR LDCHEC K,BORD ER_CON TROL", "evaluat ed_at": input.me tadata.ti mestam p, }</pre>	
OPA- H02	Hard	Passport not	Interpol SLTD	interpol. details.is		opa_res ults

(passport_stolen.registration)	flagged as stolen/lost in INTERPOL lookup	API	<pre>_stolen == true; interpol.details.is_lost == true; interpol.details.is_revoked == true; check_status == ERROR/TIMEOUT; interpol.details.is_invalid == true [Review Required]</pre>	<pre># FAILED CLOSED result := { "policy_id": "OPA-H02", "policy_name": "Passport_Stolen_Lost_Interpol", "tier": "HARD", "result": "BLOCK", "reason": "Interpol check failed or timed out —"</pre>
--------------------------------	---	-----	---	--

```
fail-  
closed  
",  
  
"data_  
source  
":  
"INTE  
RPOL_  
SLTD",  
  
"evalua  
ted_at"  
:  
input.  
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ata.tim  
estam  
p  
} if {  
  
input.e  
xternal  
_check  
s.inter  
pol.che  
ck_stat  
us in  
["ERR  
OR",  
"TIME  
OUT"]  
}  
  
#  
STOLE  
N  
else :=
```

```
{  
  
  "policy  
_id":  
  "OPA-  
H02",  
  
  "policy  
_name  
":  
  "Passp  
ort_Sto  
len_Lo  
st_Inte  
rpol",  
  
  "tier":  
  "HARD  
",  
  
  "result  
":  
  "BLOC  
K",  
  
  "reaso  
n":  
  "Passp  
ort  
flagge  
d as  
stolen  
in  
Interpo  
l SLTD  
databa  
se",
```

```
"data_
source
":
"INTE
RPOL_
SLTD",

"evalua
ted_at"
:
input.
metad
ata.tim
estam
p
} if {

input.e
xternal
_check
s.inter
pol.che
ck_stat
us ==
"BLOC
KED"

input.e
xternal
_check
s.inter
pol.det
ails.is_
stolen
==
true
}
```

```
#
LOST
else :=
{

  "policy
_id":
"OPA-
H02",

  "policy
_name
":
"Passp
ort_Sto
len_Lo
st_Inte
rpol",

  "tier":
"HARD
",

  "result
":
"BLOC
K",

  "reaso
n":
"Passp
ort
flagge
d as
lost in
Interpo
I SLTD
databa
```

```
se",

"data_
source
":
"INTE
RPOL_
SLTD",

"evalua
ted_at"
:
input.
metad
ata.tim
estam
p
} if {

input.e
xternal
_check
s.inter
pol.che
ck_stat
us ==
"BLOC
KED"

input.e
xternal
_check
s.inter
pol.det
ails.is_l
ost ==
true
}
```

```
#
REVOK
ED
else :=
{

"policy
_id":
"OPA-
H02",

"policy
_name
":
"Passp
ort_Sto
len_Lo
st_Inte
rpol",

"tier":
"HARD
",

"result
":
"BLOC
K",

"reaso
n":
"Passp
ort
flagge
d as
revoke
d in
```


Interpo
l SLTD
databa
se",

"data_
source
":
"INTE
RPOL_
SLTD",

"evalua
ted_at"
:
input.
metad
ata.tim
estam
p
} if {

input.e
xternal
_check
s.inter
pol.che
ck_stat
us ==
"BLOC
KED"

input.e
xternal
_check
s.inter
pol.det
ails.is_r

evoked

==

true

}

#

REVIE

W

else :=

{

"policy

_id":

"OPA-

H02",

"policy

_name

":

"Passp

ort_Sto

len_Lo

st_Inte

rpol",

"tier":

"HARD

",

"result

":

"REVIE

W_RE

QUIRE

D",

"reaso

n":

"Passp ort flagge d as invalid — require s casew orker review ", "data_ source ": "INTE RPOL_ SLTD", "evalua ted_at" : input. metad ata.tim estam p } if { review _requir ed } # PASS else :=
--

```
{  
  
  "policy  
_id":  
"OPA-  
H02",  
  
  "policy  
_name  
":  
"Passp  
ort_Sto  
len_Lo  
st_Inte  
rpol",  
  
  "tier":  
"HARD  
",  
  
  "result  
":  
"PASS"  
,  
  
  "reaso  
n":  
"Passp  
ort not  
found  
in  
Interpo  
I SLTD  
databa  
se",  
  
  "data_  
source
```

					<pre>": "INTE RPOL_ SLTD", "evalua ted_at" : input. metad ata.tim estam p }</pre>	
OPA- H03 (docume nt_fraud. rego)	Hard	Docume nt fraud_sc ore < 0.9 on all Tier 1 docume nts	input.do cuments [] — array of docume nt extractio n records from PostgreS QL docume nt_extra ctions table. critical_d ocument	critical_d ocument s[].fraud _score >= 0.9	result := { "policy_i d": "OPA- H03", "policy_n ame": "Docum ent_Frau d_Tier1_	opa_res ults

			<pre>_types := { "PASSPORT", "BANK_STATEMENT", "EMPLOYMENT_LETTER" , "PAYSHEET", "P60_TAX", "IELTS_CERTIFICATE", "DEGREE_CERTIFICATE" }</pre>	<pre>HardBlock", "tier": "HARD", "result": "BLOCK", "reason" : sprintf(" Tier 1 document(s) exceed fraud threshold (>= 0.9): %v", [failed_document_details]), "data_source": "DOCUMENT_E"</pre>	
--	--	--	---	--	--

XTRACT
IONS",

"evaluat
ed_at":
input.me
tadata.ti
mestam
p,

"failed_d
ocument
s":
fraudule
nt_docs,

} if {

count(fra
udulent_
docs) >
0

} else :=
{

"policy_i
d":

"OPA-
H03",

"policy_n
ame":
"Docum
ent_Frau
d_Tier1_
HardBlo
ck",

"tier":
"HARD",

"result":
"BLOCK
",

"reason"
:
sprintf("
Fraud
detectio
n
incomple
te on
Tier 1
docume
nt(s) —
fail-
closed:


```
%v",  
[incompl  
ete_doc_  
details]],
```

```
"data_so  
urce":  
"DOCU  
MENT_E  
XTRACT  
IONS",
```

```
"evaluat  
ed_at":  
input.me  
tadata.ti  
mestam  
p,
```

```
} if {
```

```
count(in  
complet  
e_fraud_  
docs) >  
0
```

```
} else :=  
{
```

"policy_id":
"OPA-H03",

"policy_name":
"Document_Fraud_Tier1_HardBlock",

"tier":
"HARD",

"result":
"PASS",

"reason":
:
sprintf("All %d Tier 1 documents passed fraud

					<pre> detectio n (all scores < 0.9)", [count(ti er_1_doc s)]), "data_so urce": "DOCU MENT_E XTRACT IONS", "evaluat ed_at": input.me tadata.ti mestam p, } </pre>	
OPA- H05 (data_re sidency.r ego)	Hard	Data residenc y — all PII stored only in europe- west2 or	Raise question in query log Q29			opa_res ults

		europe-west1				
OPA-H06 (document_tampering.rego)	Hard	Blocks applications with strong evidence of document forgery or tampering	Passport Verification API Stored in Postgres document_extractions.raw_extraction (JSONB) → Normalized into normalized_fields	mrz_cross_reference.overall_match == false; document_status.is_cancelled == true; authenticity_authenticity_score < 0.50; check_status == ERROR/TIMEOUT Field details from extraction to be mapped - TBD	# FAIL-CLOSED result := { "policy_id": "OPA-H06", "policy_name": "Document_Tampering_Check", "tier": "HARD", "result": "BLOCK", "reason": "Passport verification failed or timed out — fail-closed", "data_source": "OPA-H06"	opa_results

urce":
"PASSP
ORT_VE
RIFY_AP
I",

"evaluat
ed_at":
input.me
tadata.ti
mestam
p
} if {

input.ext
ernal_ch
ecks.pas
sport_ve
rify.chec
k_status
in
["ERROR
",
"TIMEO
UT"]
}

MRZ
MISMAT
CH
else := {

"policy_i
d":
"OPA-
H06",

```
"policy_name":
"Document_Tampering_Check",
  "tier":
"HARD",

"result":
"BLOCK",

"reason":
"MRZ mismatch detected in passport",

"data_source":
"PASSPORT_VERIFY_API",

"evaluated_at":
input.metadata.timestamp
} if {
```

```
input.external_checks.pasport_verify.check_status == "BLOCKED"
```

```
input.external_checks.pasport_verify.details.mrz_cross_reference.overall_match == false  
}
```

```
# LOW AUTHENTICITY  
else := {
```

```
"policy_id":  
"OPA-H06",
```

```
"policy_name":
```

						<pre>"Document_Tampering_Check", "tier": "HARD", "result": "BLOCK ", "reason" : "Passport authenticity score below threshold", "data_source": "PASSPORT_VERIFY_API", "evaluated_at": input.metadata.timestamp } if {</pre>	
--	--	--	--	--	--	--	--


```
input.external_checks.pas  
sport_verify.check_status  
==  
"BLOCKED"
```

```
input.external_checks.pas  
sport_verify.details.authenticity.authenticity_score <  
0.50  
}
```

```
#  
CANCELED
```

```
else := {
```

```
"policy_id":  
"OPA-H06",
```

```
"policy_name":  
"Document_Tam
```

						pering_C heck", "tier": "HARD", "result": "BLOCK ", "reason" : "Passpor t marked as cancelle d", "data_so urce": "PASSP ORT_VE RIFY_AP I", "evaluat ed_at": input.me tadata.ti mestam p } if { input.ext ernal_ch ecks.pas sport_ve rify.chec	
--	--	--	--	--	--	--	--

```
k_status
```

```
==
```

```
"BLOCK  
ED"
```

```
input.ext  
ernal_ch  
ecks.pas  
sport_ve  
rify.detail  
s.docum  
ent_stat  
us.is_ca  
ncelled  
== true  
}
```

```
# PASS
```

```
else := {
```

```
"policy_i  
d":
```

```
"OPA-  
H06",
```

```
"policy_n  
ame":
```

```
"Docum  
ent_Tam  
pering_C  
heck",
```

```
"tier":  
"HARD",
```

```
"result":
```

```
"PASS",
```

					<pre>"reason": : "Passport verification passed", "data_source": "PASSPORT_VERIFY_API", "evaluated_at": input.metadata.timestamp }</pre>	
OPA-S01 (biometric_borderline.rego)	Soft	Biometric match score between 0.75-0.85 (borderline — human review)	Json Payload(Visakey)	TBD	<pre># FLAG FOR REVIEW result := { "policy_id": "OPA-S01", "policy_n</pre>	opa_results

ame": "Biometr ic_Borde rline_Ch eck", "tier": "SOFT", "result": "FLAG_F OR_REV IEW", "reason" : "Borderli ne biometri c/authen ticity score detected ", "data_so urce": "PASSP ORT_VE RIFY_AP I", "evaluat ed_at": input.me tadata.ti mestam p	
---	--

```
} if {  
  
    flag_for_  
    review  
}  
  
# FAIL-  
OPEN  
(IMPOR  
TANT)  
else := {  
  
    "policy_i  
    d":  
    "OPA-  
    S01",  
  
    "policy_n  
    ame":  
    "Biometr  
    ic_Borde  
    rline_Ch  
    eck",  
    "tier":  
    "SOFT",  
  
    "result":  
    "PASS",  
  
    "reason"  
    :  
    "Passpor  
    t  
    verificati  
    on  
    unavaila
```

```
ble —
fail-
open",

"data_so
urce":
"PASSP
ORT_VE
RIFY_AP
I",

"evaluat
ed_at":
input.me
tadata.ti
mestam
p
} if {

input.ext
ernal_ch
ecks.pas
sport_ve
rify.chec
k_status
in
["ERROR
",
"TIMEO
UT"]
}

# PASS
else := {

"policy_i
```

d":	"OPA-S01",
"policy_name":	"Biometric_Borderline_Check",
"tier":	"SOFT",
"result":	"PASS",
"reason":	"Biometric/authenticity score acceptable",
"data_source":	"PASSPORT_VERIFY_API",
"evaluated_at":	input.metadata.timestamp

					p }	
OPA-S03 (completeness_low.rego)	Soft	Document completeness_score 60-70%	input.documents [] — array of document extraction records from PostgreSQL document_extractions table.	Completeness score 60-70%	TBD	opa_results
OPA-S04 (rapid_submission.rego)	Soft	Application submitted within 3 hours (potential bot activity) Flags applications with risky submission behaviour or suspicion	Awaiting Clarification			opa_results

		us contact details				
OPA-S05 (enhanced_scrutiny.rego)	Soft	Nationality flagged for enhanced scrutiny (not blocked, but heightened)	Awaiting clarification			opa_results
OPA-S06 (cmek_rotation.rego)	Soft	CMEK key rotation overdue (infrastructure compliance alert) Flags applications with adverse border control or travel history	Awaiting Clarification			opa_results